Rethinking Post-Hoc Search-Based Neural Approaches for Solving Large-Scale Traveling Salesman Problems

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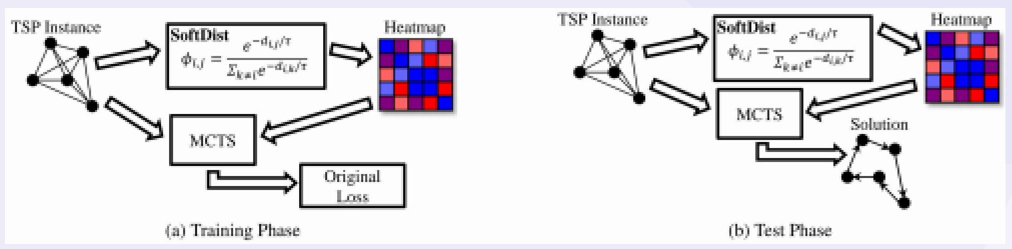
01 Introduction: Heatmap-Guided MCTS—Questions and Contributions.

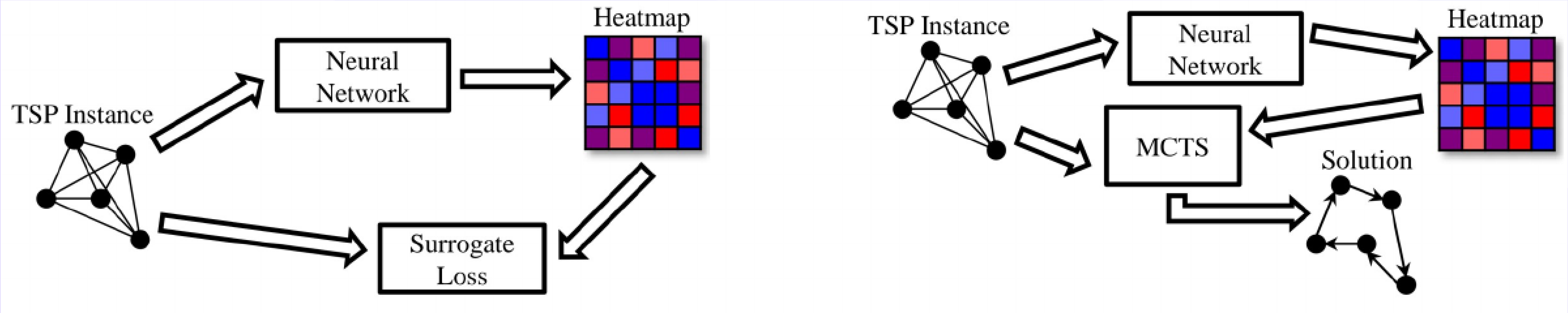
- ML heatmaps + MCTS face train–test mismatch and scaling limits.
- SoftDist provides a supervision-free heatmap tuned with MCTS for aligned objectives.
- A Score metric reveals persistent inefficiency of MCTS versus **LKH-3** under equal compute.
- Experiments show SoftDist matches or surpasses most **ML heatmaps** with lower latency.
- Positioning: prior heatmaps train without MCTS; we address this misalignment.

02 SoftDist: Simple Distance-Based Heatmap and MCTS-Aligned Tuning



- Heatmap: softmax on distances with one **temperature** τ.
- Tune τ by grid search with MCTS-in-the-loop to minimize tour length.
- **Label-free**, fast, and training–inference aligned.

03 Preliminaries: TSP and How MCTS Consumes Heatmaps



(a) Training Phase (b) Test Phase

- 2D Euclidean **TSP**: minimize tour length.
- Models output an n×n **heatmap** guiding choices.
- Within **MCTS**: construct tour, sample guided moves, apply k-opt.
- Backpropagate improvements; our focus is **inference-time** use of heatmaps.

04 Evaluation Protocol: Fair LKH-**3** Head-to-Head, Score Metric, Datasets, and Settings

- A compute-equal head-to-head matches wall-clock for MCTS and LKH-3.
- The **Score** equals the gap of LKH-3 divided by the gap of MCTS, with higher favoring MCTS.
- **Datasets** include TSP 500, TSP 1000, and TSP 10000 drawn from public benchmarks.

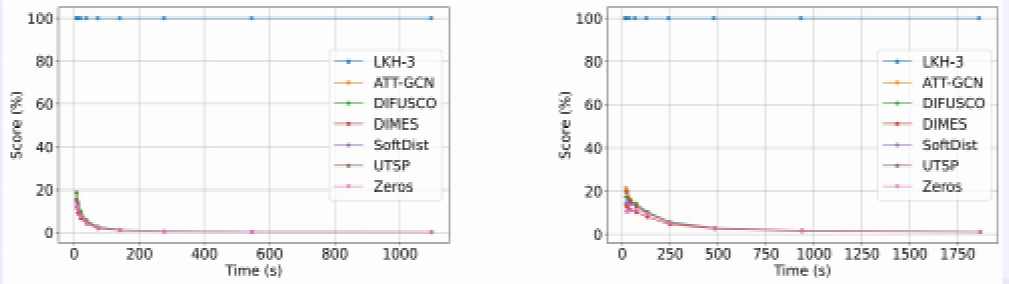
- **Tuning**: Temperature values are tuned by grid search on held out sets with MCTS in the loop.
- **Metrics**: tour length, gap, total time, and the Score under matched budgets.

05 Results: SoftDist vs ML Heatmaps; MCTS vs LKH-**3** via Score

- SoftDist matches or surpasses most machine learned heatmaps with minimal latency.
- A diffusion based method is competitive but computationally costly, while SoftDist is **label-free**.
- The Score shows MCTS far below **LKH-3** under equal compute across datasets.
- On large instances, SoftDist remains Pareto competitive in quality versus latency.
- Reported gaps are small for SoftDist but remain above **LKH-3**.
- SoftDist gaps: **1.44%/2.20%/3.13%** (500/1000/10000).

06 Sensitivity: MCTS Settings and Time Budgets

- **Tuned** MCTS settings improve most methods yet do not close the gap to **LKH-3**.
- With more time, LKH-3 approaches optimal solutions and MCTS Scores decrease.
- On **TSP-10000**, SoftDist performs best among MCTS variants yet remains below **LKH-3**.
- Sensitivity indicates reduced **heatmap influence** under stronger MCTS tuning.
- Time budget curves **plateau** for large instances, showing limited MCTS returns.



(a) TSP-500 with different MCTS settings. (b) TSP-1000 with different MCTS settings.

Method	TSP-500 Length ↓	Gap ↓	Time ↓	Score ↑	TSP-1000 Length ↓	Gap ↓	Time ↓	Score ↑
LKH-3 (default)	16.55*	0.00%	46.28m	—	23.12*	0.00%	2.57h	—
ATT-GCN†	16.72	1.02%	0.52m+0.67m	5.38%	23.48	1.58%	0.73m+1.43m	13.77%
DIMES†	16.75	1.26%	0.97m+0.68m	4.35%	23.61	2.11%	2.08m+1.45m	10.29%
DIFUSCO#	16.69	0.90%	3.61m+0.68m	6.12%	23.48	1.56%	11.86m+1.43m	13.93%
UTSP†	16.73	1.09%	1.37m+0.68m	5.05%	23.50	1.65%	3.35m+1.45m	13.18%
SoftDist	16.72	1.03%	0.00m+0.68m	5.32%	23.52	1.73%	0.00m+1.44m	12.56%
Zeros	16.72	1.06%	0.00m+0.68m	5.20%	23.55	1.85%	0.00m+1.44m	11.72%

07 Conclusion and Outlook: Rethinking Heatmap-Guided TSP Solving

- SoftDist is competitive in quality and fast, while MCTS trails **LKH-3** by the Score metric.
- Misaligned surrogate training undermines practicality of heatmap guided search.
- Future directions include theory grounded objectives and **end to end** solvers.

- The evidence questions the practical value of **post hoc** neural search for large scale TSP.
- We advocate simpler baselines and fair compute matched comparisons in future studies.

Thank You

Questions & Discussion